Software Requirements Specification

for

FACULTY APPOINTMENT MANAGEMENT SYSTEM

Version 1.0

Prepared by

Group 4

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1. Introduction

1.1 Purpose

This document specifies the Software Requirement Specification (**SRS**) of the Faculty Appointment Management System (**FAMS**). The purpose of the system is to provide a centralized, secure, and efficient platform for students of NIT Calicut to request appointments with faculty members and for faculty to manage their availability and appointments directly.

1.2 Document Conventions

- **Shall** indicates a mandatory requirement.
- **Should** indicate a recommended feature.
- **May** indicates an optional feature.
- Roles are denoted as **Student**, **Faculty**, and **Admin**.
- Time values are represented in 24-hour IST format.

1.3 Intended Audience and Reading Suggestions

- **Students & Faculty Evaluators:** Review functional completeness and correctness
- **Developers:** Use sections 3 and 4 for implementation guidance
- **Testers:** Use sections 4 and 5 for validation and verification
- **Admins:** Refer to the admin panel and the Business Rules section

1.4 Product Scope

FAMS aims to reduce the overhead and uncertainty involved in meeting faculty members by digitalising appointment scheduling. The system improves accessibility, minimises scheduling conflicts, and provides analytics for institutional oversight.

1.5 References

- IEEE 830 / IEEE SRS Standard
- NIT Calicut Academic Regulation
- Google Calendar API Documentation

2. Overall Description

2.1 Product Perspective

The FAMS is a mobile-based application integrated with authentication systems and an optional third-party calendar service.

2.2 Product Functions

- Institutional authentication and RBAC
- Faculty availability and slot management
- Student appointment requests
- Approval, rejection, and cancellation workflows
- Email and push notifications
- Admin analytics and configuration
- Timetable-based schedule population via a CSV / JSON file

2.3 User Classes and Characteristics

User Class	Description
Student	UG, PG, PhD students who are part of NITC
Faculty	Teaching staff managing schedules and student interactions
Admin	Department or Institute-level administrators

2.4 Operating Environment

- Client: Android OS
- Frontend: Next.Js
- Backend: Python Fast API
- Server OS: Linux-based cloud or on-premise server
- Database: PostgreSQL

2.5 Design and Implementation Constraint

- Only institute email IDs (@nitc.ac.in) are allowed
- Compliance with institute's data policies
- Internet connectivity is required

2.6 User Documentation

- Online user manual
- Quick-start guide for students and faculties
- Admin configuration guide

2.7 Assumptions and Dependencies

- Institute email directory access is available
- Faculty cooperation in schedule confirmation
- Email service availability

3. External Interface Requirements

3.1 User Interfaces

- Android mobile application built using Next.js
- Faculty dashboard with calendar and list views optimised for mobile
- Student booking interface visual slot indicators
- Admin panel with analytics and charts

3.2 Hardware Interfaces

- No special hardware is needed

3.3 Software Interfaces

- Institutional OAuth / SSO
- Google Calendar APIs
- Email service
- PostgreSQL
- Python FASTAPI REST APIs

3.4 Communication Interfaces

- HTTPS for all client-server communication
- REST-based APIs for backend services

4. System Features

4.1 User Authentication & Role Management

4.1.1 Description and Priority

This feature provides secure authentication using institutional credentials and enforces role-based access control among Students, Faculty, PhD Scholars, and Admins. It ensures that only authorized NIT Calicut users can access the system.

Priority: High

Benefit: 9 | **Penalty:** 9

4.1.2 Stimulus/Response Sequences

1. The user opens the android application.
2. User selects “Login” and enters institutional email credentials.
3. System verifies credentials via institutional authentication service.
4. The system identifies the user role (Student / Faculty / Admin).
5. The system redirects the user to the appropriate dashboard.

4.1.3 Functional Requirements

- **REQ-1:** The system shall allow login only via institutional email IDs
- **REQ-2:** The system shall authenticate users using secure institutional authentication mechanisms
- **REQ-3:** The system shall assign exactly one primary role to each user.
- **REQ-4:** The system shall restrict access to features based on assigned roles.
- **REQ-5:** The system shall display an appropriate error message for invalid login attempts.

4.2 Faculty Dashboard

4.2.1 Description and Priority

The Faculty Dashboard provides a centralized interface for managing student meetings. A key component of this feature is the Approval Workflow, ensuring that no appointment is officially scheduled until the faculty member reviews and accepts the request.
Priority: High
Benefit: 8 | **Penalty:** 8

4.2.2 Stimulus/Response Sequences

- **Stimulus:** Faculty views a "Pending" request on the calendar.
- **Response:** The system provides options to "Accept" or "Cancel/Decline."
- **Stimulus:** Faculty clicks "Accept."
- **Response:** The system updates the status to "Approved," moves the appointment to the scheduled list, and notifies the student.
- **Stimulus:** Faculty clicks "Cancel/Decline."
- **Response:** The system opens a feedback box and, upon submission, removes the request and notifies the student.

4.2.3 Functional Requirements

- **REQ-1: Calendar Visualisation The system shall display a calendar-like interface that shows** all appointment requests. Requests must be visually categorised into two states: **Pending Approval** and **Approved/Scheduled**.
- **REQ-2: Appointment Acceptance** The system shall provide a mechanism for faculty to "Accept" a pending appointment request. Only upon acceptance shall the appointment be officially blocked off in the system and the student notified.
- **REQ-3: Appointment Cancellation and Feedback**. The system shall allow faculty members to cancel/decline any appointment (whether pending or approved). The system must require the faculty to enter a reason in a comment box before the cancellation is finalised.
- **REQ-4:** Faculty shall enable or disable availability using Busy Mode.
- **REQ-5**: PhD scholars shall manage availability only under supervising faculty control.

4.3 Student Dashboard

4.3.1 Search & Discovery

4.3.1.1 Description and Priority

This feature allows students to locate specific faculty members or find available staff within a department.
Priority: High
Benefit: 8 | **Penalty**: 8

4.3.1.2 Stimulus/Response Sequences

- **Stimulus**: Student enters text into the universal search bar.
  Response: System filters and displays faculty profiles matching the name, department, or keyword.

4.3.1.3 Functional Requirements

- **REQ-1:** The system shall provide a single-field search interface to query faculty by Name, Department, and Research Keywords simultaneously.
- **REQ-2:** The system shall display search results dynamically, refreshing the list adequately.
- **REQ-3:** If no matches are found, the system shall display a "No Faculty Found" message with a suggestion to clear active filters.

4.3.2 Appointment Requesting (Queue System)

4.3.2.1 Description and Priority: Allows students to apply for specific time slots. This follows a "request-and-approve" model where multiple students can apply for the same slot, forming a queue for the faculty to review.

Priority: High
Benefit: 9 | **Penalty**: 9

4.3.2.2 Stimulus/Response Sequences

- **Stimulus**: Students select an "Open Slot" from a faculty's calendar.
  Response: System opens a booking form requiring "Purpose" and "Description."
- **Stimulus**: Student submits the booking form.
  Response: System records the request, marks the status as "Pending" in the student's dashboard, and adds the student to the faculty's applicant queue for that slot.

4.3.2.3 Functional Requirements

- **REQ-5:** The system shall allow students to submit a request for a slot even if other students have already applied for the same time.
- **REQ-6**: The system shall enforce that the "Meeting Purpose" field is not empty and contains at least 10 characters.
- **REQ-7**: The system shall validate that the "Description" field is not empty and contains at least 10 characters.
- **REQ-8**: The system shall provide a "My Requests" view where students can track the real-time status (Pending, Confirmed, Declined) of all submissions.
- **REQ-9**: Upon a faculty member confirming a different student for a slot, the system shall automatically update all other applicants for that specific slot to "Declined."

4.4 Appointment Workflow & Logic

4.4.1 Description and Priority

This feature governs the complete lifecycle of an appointment between a student and a faculty member, including booking, validation, rescheduling, cancellation, and no-show handling.
It ensures fair access to faculty time, prevents misuse of the booking system
Priority: High
Benefit: 7 | **Penalty:** 7

4.4.2 Stimulus/Response Sequences

- **Stimulus:** Student attempts to book a slot less than the configured booking window (e.g., 1 hour before start time).
  Response: System disables the slot and displays a tooltip indicating that the booking window has closed.
- **Stimulus:** Faculty member requests a reschedule for an already confirmed appointment.
  Response: System changes the appointment status to *"Reschedule Requested"* and notifies the student with the proposed new time.
- **Stimulus:** Student cancels an appointment within the allowed cancellation window.
  Response: System marks the appointment as *"Cancelled by Student"* and releases the slot back to faculty availability.
- **Stimulus:** Faculty marks a student as a "No-Show."
  Response: System records the no-show event, increments the student's no-show counter, and flags the account if the threshold is exceeded.
- **Stimulus:** Student exceeds the allowed number of no-shows.
  Response: System restricts further bookings and notifies the Admin for review and potential suspension.

4.4.3 Functional Requirements

- **REQ-1:** The system shall enforce a configurable booking window such that appointments cannot be booked less than *X* hours before the scheduled start time.
- **REQ-2:** The system shall allow faculty members to propose a new time for an existing appointment instead of canceling it outright.
- **REQ-4:** The system shall allow students to cancel appointments only if the cancellation occurs at least *X* hours before the appointment time.
- **REQ-5:** The system shall allow faculty members to mark an appointment as *"No-Show"* after the scheduled time has passed.
- **REQ-6:** The system shall maintain a count of no-show incidents per student account.
- **REQ-7:** If a student exceeds the configured no-show threshold, the system shall restrict further appointment bookings pending Admin intervention.
- **REQ-8:** The system shall notify Administrators when a student account is flagged due to excessive no-shows.

4.5 Notification & Alerts

4.5.1 Description and Priority

This feature ensures timely communication between students, faculty, PhD scholars, and administrators by delivering notifications and alerts for important appointment-related events. Notifications help reduce missed appointments and improve coordination.
Priority: High
Benefit: 8 | **Penalty:** 7

4.5.2 Stimulus/Response Sequences

1. The student submits an appointment request.
2. The system triggers a notification to the concerned faculty or PhD scholar.
3. Faculty approves, rejects, or reschedules the appointment.
4. The system sends a corresponding notification to the student.
5. Prior to the appointment time, the system sends reminder notifications.

4.5.3 Functional Requirements

- **REQ-1:** The system shall send an email notification when an appointment request is created.
- **REQ-2:** The system shall notify students when an appointment is approved, rejected, or rescheduled.
- **REQ-3:** The system shall notify faculty and PhD scholars of new appointment requests.

- **REQ-4:** The system shall send reminder notifications 1 hour before an approved appointment.
- **REQ-5:** The system shall generate a daily summary notification for faculty listing the day's appointments.
- **REQ-6:** The system shall send weekly schedule update reminders to faculty members.
- **REQ-7:** The system shall add an "Add to Calendar" link in the notification mail.

4.6 Admin Panel

4.6.1 Description and Priority

The Admin Panel provides system administrators with centralised tools for platform maintenance, governance, and oversight.
It enables secure user lifecycle management, institutional structure configuration, and analytics-driven monitoring to ensure system integrity and continuous improvement.

Priority: Medium–High
Benefit: 9 | **Penalty:** 9

4.6.2 Stimulus/Response Sequences

- **Stimulus:** Administrator creates a new faculty account.
  Response: System validates required fields, creates the account, and sends onboarding credentials to the faculty member.
- **Stimulus:** Administrator deactivates a student account marked as alumni. (csv format)
  Response: System disables login access and prevents the account from booking or managing appointments.
- **Stimulus:** Administrator uploads a CSV or JSON file containing faculty timetable data.
  Response: System validates the file format, parses timetable entries, and maps busy slots to faculty calendars.
- **Stimulus:** Administrator adds a new academic department.
  Response: System registers the department and makes it available for faculty assignment and student search filters.
- **Stimulus:** Administrator modifies a department's office location.
  Response: System updates the department record and reflects the change across faculty profiles and booking views.
- **Stimulus:** Administrator opens the analytics dashboard.
  Response: System aggregates and displays usage metrics, including appointment volume and faculty response times.

4.6.3 Functional Requirements

- **REQ-1:** The system shall provide administrators with a secure interface to create, activate, deactivate, or delete user accounts.
- **REQ-2:** The system shall allow administrators to onboard new faculty members by assigning departments, office locations, and default availability settings.
- **REQ-3:** The system shall allow administrators to deactivate student accounts identified as alumni or inactive users.
- **REQ-4:** The system shall allow administrators to add, remove, or edit academic departments.
- **REQ-5:** The system shall allow administrators to configure or update office locations associated with departments or faculty profiles.
- **REQ-6:** The system shall provide an analytics dashboard displaying appointment usage statistics grouped by department.
- **REQ-7:** The system shall calculate and display the average faculty response time from booking request to confirmation.
- **REQ-8:** The system shall allow administrators to export analytics data in a standard format (CSV or PDF).
- **REQ-9:** The system shall allow administrators to upload faculty timetables in CSV or JSON format and automatically mark time slots corresponding to classes or busy periods as unavailable for booking.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall support concurrent appointment requests.
- Page response time shall be minimised.

5.2 Safety Requirements

- No critical safety risks identified.

5.3 Security Requirements

- All data shall be transmitted over HTTPS.
- Passwords shall be securely salted.
- Access control shall be strictly enforced

5.4 Software Quality Attributes

- **Usability:** Mobile-first, intuitive Android UI
- **Reliability:** Minimal downtime during academic hours
- **Scalability:** Support growth in users and departments
- **Maintainability:** Modular Spring Boot backend and clean Flutter architecture

- **Postability:** Backend deployable across Linux environments

5.5 Business Rules

- Only institute members may access the system.
- Faculty decisions on appointments are final.

6. Other Requirements

- Audit logs for admin actions
- Data backup and recovery mechanisms

Appendix A: Glossary

- **Slot:** A time interval available for appointments
- **No-Show:** Student failing to attend an approved appointment
- **RBAC:** Role-Based Access Control

Appendix B: Analysis Models

- Use Case Diagrams
- Sequence Diagrams
- ER Diagrams